

Medical School Environments and Physician Treatment Styles

SUPPLEMENTAL APPENDIX

Shirley Cai* Ian McCarthy†

May 11, 2026

A Sample Construction Details

This section provides additional detail on the construction of our cardiologist-year analytical panel, the medical school-to-HRR crosswalk, the AHA training-period cath lab measure, and the NIH ExPORTER ranking panel.

A.1 Cardiologist Assignment and Volume Restriction

For each NSTEMI episode, we assign a single attending cardiologist following the procedure of [Molitor \(2018\)](#). The hierarchy prioritizes the first cardiologist who delivers a consultation (CPT codes 99251–99255 prior to their elimination by CMS in January 2010, with the hierarchy collapsing to first cardiologist of record for inpatient care for episodes thereafter), the first cardiologist of record on the inpatient stay, and the first cardiologist seen by the patient in carrier-line claims within 90 days of the index admission. We exclude episodes for which no cardiologist could be assigned and patient-years for which Medicare fee-for-service coverage was not present for the entire year. Cardiologists are identified in MD-PPAS using primary-specialty fields for cardiology, interventional cardiology, clinical cardiac electrophysiology, and advanced heart failure and transplant cardiology.

We restrict to cardiologist-years with at least 11 NSTEMI patients. The threshold provides stable per-physician estimates of catheterization rates and aligns with the CMS cell-size

*Emory University.

†Emory University. ian.mccarthy@emory.edu

threshold for public-use files. A trade-off of the volume restriction is that established, higher-volume cardiologists are over-represented in our analytical sample relative to the broader cardiologist population.

A.2 Residualized Catheterization Rate

Our outcome is the volume-weighted mean residualized cath-within-two-days indicator at the cardiologist-year level. At the patient level, we estimate a linear probability model regressing the cath-within-two-days indicator on the 31 Elixhauser comorbidity flags and year fixed effects, retaining the residual as the patient-level outcome. We then aggregate to the cardiologist-year by taking the volume-weighted mean residual. The residualization removes mechanical variation across cardiologists that would arise from differences in patient comorbidity mix.

Elixhauser comorbidities are constructed from a lookup table of approximately 2,600 ICD-9 and ICD-10 diagnosis codes mapped to 31 categories, drawn from the cardiologist’s NSTEMI-episode and 12-month-look-back claims. A given diagnosis code may map to multiple Elixhauser categories, so the lookup is implemented as a join rather than a categorical recoding.

A.3 Medical School-to-HRR Crosswalk

The medical school HRR is constructed in three stages. First, we obtain each cardiologist’s medical school name and graduation year from the CMS Physician Compare National Downloadable File. Second, we map medical school names to ZIP codes using a hand-curated crosswalk restricted to LCME-accredited U.S. allopathic medical schools, validated against the LCME accreditation directory. Of 187 medical schools in our crosswalk, 186 resolve via the LCME accreditation join, and one historically defunct school resolves via a separate fallback ZIP. Schools coded in Physician Compare as “OTHER” are excluded; the OTHER category captures most foreign-trained physicians and a small number of unmatched U.S. schools. Third, we map ZIP codes to hospital referral regions using the Dartmouth Atlas ZIP-to-HRR crosswalk (vintage 2015).

Our analytical panel restricts to allopathic MD graduates because only roughly 11% of U.S. medical school graduates are osteopathic (DO), with an even smaller share among cardiologists (AAMC, U.S. Physician Workforce Data Dashboard).

A.4 AHA Training-Period Cath Lab Measure

Cath lab availability is recorded in the AHA Annual Survey CCLABHOS variable from 1994 onward and in the predecessor CCLAB82 variable from 1980–1993. Both are recoded to a binary indicator of direct hospital provision of cardiac catheterization. The HRR-year cath lab share is then computed as the share of AHA-respondent hospitals in each HRR-year for which the binary indicator equals one. The aggregated AHA series exhibits a smooth secular trend across the 1994 questionnaire revision, with the national HRR-mean cath lab share rising from approximately 0.14 in 1980 to approximately 0.39 in 2003. We interpret the smooth trend as evidence that the recoding from CCLAB82 to CCLABHOS does not introduce a measurement break.

For each cardiologist, we match the AHA HRR-year cath lab share at the year the cardiologist began medical school, defined as graduation year minus three (the modal duration of allopathic medical school in the United States). AHA cath lab data are available 1980–2003, so the year-matching produces clean assignments for cardiologists with graduation years 1983 through 2006. We restrict our analytical sample to this graduation-year window.

A.5 Year-Matching and Cohort Variation

Our preferred specification identifies β_{train} off variation in cath lab availability across graduation cohorts who attended medical school in the same HRR. The variation comes from two sources. First, the rollout of cath labs occurred at different times across HRRs over the 1980–2003 period, so the same medical school HRR exposed different cohorts to different cath lab shares. Second, even within HRRs that had cath labs early, the share of hospitals offering cath services continued to rise across the rollout period. We rely on both sources of within-HRR temporal variation in our identification.

A.6 NIH ExPORTER Ranking Panel

To distinguish procedural exposure from medical school prestige, we construct a year-matched panel of medical school NIH research funding from the NIH ExPORTER bulk files. We aggregate annual project-level award totals (1985–2025) to the medical school level using the institution-name field, with consolidation of name variants (abbreviations, mergers, renamings) by hand. Each medical school then receives a within-year rank in total NIH funding. The constructed series is validated against the published Blue Ridge Institute for Medical Research rankings, with a Pearson correlation across overlapping years above 0.99.

For each cardiologist, we match the medical school NIH rank in the year of medical school

start (graduation year minus three), parallel to the AHA cath lab matching procedure. We use both continuous $-\log(\text{rank})$ and tier indicators (Top 10, Top 11–25, Top 26–50, Top 51–100, with unranked schools as the reference) in our specifications.

A.7 GME Duration Filter

We drop physician-years for which the implied training pipeline is incompatible with the cardiologist’s observed graduation year and subspecialty. The required GME duration after medical school is six years for general cardiology, seven years for interventional cardiology, eight years for clinical cardiac electrophysiology, and seven years for advanced heart failure. Physician-years implying a training pipeline shorter than the GME duration are dropped. The filter removes approximately 70 of approximately 31,000 physician-year observations.

B Cross-Sectional Balance, by Subspecialty

Table B.1 reports the training-period cath lab comparison between cross-sectional movers and stayers separately within each cardiology subspecialty. The ordering is similar across subspecialties, with movers in each subspecialty coming from medical school HRRs with lower cath lab share than stayers in that same subspecialty. The differences are statistically significant in general cardiology, interventional cardiology, and electrophysiology. Advanced heart failure has too few stayers (six cardiologists) to support a precise within-subspecialty test.

Table B.1. Cross-sectional balance by cardiology subspecialty

Subspecialty	Stayers		Movers		Diff.	p
	N	Train. cath	N	Train. cath		
General Cardiology	491	0.034	3,075	0.021	−0.013	<0.001
Interventional	136	0.042	875	0.027	−0.015	<0.001
Electrophysiology	23	0.041	173	0.021	−0.020	0.007
Adv. Heart Failure	6	0.022	26	0.023	0.001	0.974

C Head-to-Head with Current Peer Environment

The within-origin specification in the main paper absorbs the current peer environment through destination HRR fixed effects, which condition out all cross-HRR variation in peer

cath rates. A natural alternative is to replace the destination HRR fixed effects with a continuous control for current peer cath culture, which allows training-period exposure and current peer environment to be compared on the same scale. Table C.2 reports the head-to-head specification, in which both the training-period cath lab share and the leave-one-out current-HRR peer cath rate (z -scored) enter as regressors, with year fixed effects and clustering at the medical school HRR level.

Table C.2. Head-to-head specification, training cath share and current peer cath culture

	Training only	Current only	Both
Training-HRR cath lab share (z)	0.006*** (0.002)		0.006*** (0.002)
Current-HRR cath culture (z)		0.004 (0.002)	0.004 (0.002)
Practice HRR FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	10,729	10,729	10,729

The training-period cath lab share retains a statistically significant marginal effect of 0.006 (SE 0.002, $p < 0.001$), while the current-HRR peer cath culture coefficient is smaller (0.004, SE 0.002) and not statistically distinguishable from zero at the 5% level. A cardiologist who trained in a procedurally rich market continues to catheterize at higher rates than one who trained in a procedurally poor market, even when current peer environment is included as a continuous covariate rather than absorbed by HRR fixed effects.

D NIH Rank Specifications

The main paper documents that medical school NIH-funding rank does not predict later catheterization rates, in contrast to the cath lab availability measure, which does. This section reports additional rank specifications that test the robustness of the rank null result.

The continuous $-\log(\text{rank})$ specification, the binary Top-25 indicator, and the four-tier specification all yield small and statistically insignificant coefficients on the cardiologist-year panel. Restricting to the mover sample produces similar nulls. The interpretation is that whatever channel rank captures, it does not show up in NSTEMI cath rates. The rank-by-cath stratification reported in the main paper further indicates that the cath lab imprint is similar in magnitude within the Top-25 and Below-Top-25 subsamples, confirming that cath lab availability and rank capture nearly orthogonal dimensions of the medical school environment.

E Selection Diagnostics

The main analysis addresses two distinct selection concerns. The first is that movers and stayers differ on observable characteristics, raising the possibility that the imprint estimate reflects mover-vs-stayer selection rather than a causal training effect. The second is that cardiologists from cath-rich training environments may systematically end up in cath-rich practice destinations, in which case our cross-sectional identification would conflate training imprint with destination-type sorting. We address the first with the inverse-probability-weighted (IPW) specification reported in the main paper. This section reports a parallel IPW exercise using a complementary measure of training-environment intensity, and a separate diagnostic for the destination-sorting concern.

E.1 IPW Robustness Using the Peer-Cath Measure

The main paper reports the IPW-weighted within-origin specification on the AHA training-period cath lab measure. Table E.3 reports a parallel IPW exercise using the leave-one-out peer cath rate in the medical school HRR as a complementary measure of training-environment intensity. The IPW weighting moves the imprint coefficient modestly relative to the unweighted within-origin baseline, in the same direction and of comparable magnitude to the shift we observe with the AHA measure in the main paper. Both exercises indicate that observable selection on graduation cohort, gender, subspecialty, and training-period intensity does not drive the imprint we estimate.

Table E.3. Selection robustness via inverse probability weighting (peer-cath measure)

	Baseline	IPW
Med school HRR intensity	0.076 (0.060)	0.087 (0.059)
Δ intensity	0.030 (0.028)	0.025 (0.028)
Practice HRR FE	Yes	Yes
Year FE	Yes	Yes
IPW	No	Yes
Observations	13,174	13,174

E.2 Training Intensity and Destination Choice

A separate identification concern is that cardiologists from cath-rich training environments may end up in cath-rich current peer environments. If so, the training imprint we estimate in the main paper could partly reflect destination-type sorting on the training measure rather than a causal effect of training. To assess this directly, we regress each cardiologist’s current peer cath rate (the leave-one-out residualized cath rate of other cardiologists in the same HRR-year) on their training-period cath lab share, with year fixed effects and clustering at the medical school HRR level.

On the full cardiologist-year panel (10,736 observations from 3,207 cardiologists with non-missing training-period and current-peer data), the coefficient on training-period cath lab share is -0.012 (SE 0.016), statistically indistinguishable from zero. Restricting to each cardiologist’s earliest panel observation as a cross-section produces a similarly small estimate: -0.027 (SE 0.018), $n = 3,207$. The pairwise correlation between training-period cath lab share and current peer cath rate is -0.03 .

Training-period cath lab availability does not predict the current peer cath rate in our sample. Cardiologists from cath-rich training environments are not systematically located in cath-rich destinations, and the destination-type-sorting concern does not appear to drive our cross-sectional imprint estimate. We caution that our cardiologist-level cross-section uses the earliest panel observation rather than the first practice location after training, so this is a contemporaneous diagnostic rather than a direct test of training-induced sorting at first placement.

F Event-Study Details

This section reports the full coefficient table for the within-physician event study around mid-career moves.

Table F.4 reports point estimates, standard errors, and 95% confidence intervals for each event-time coefficient $\hat{\beta}_\tau$ in the within-physician event study, with $\tau \in [-3, 5]$ and $\tau = -1$ omitted as the reference period. The pre-move coefficients ($\tau \in \{-3, -2\}$) are statistically indistinguishable from zero, consistent with the parallel-trends assumption. Adaptation jumps at $\tau = 0$ to 0.443 (SE 0.097, 95% CI [0.25, 0.63]) and persists through $\tau = 5$. The $\tau \in \{2, 3\}$ coefficients are smaller and noisier than the $\tau = 0, 1, 4, 5$ coefficients, but the post-move trajectory is broadly stable rather than fading.

Pre-trends are formally tested by the joint F -statistic on $\hat{\beta}_{-3}$ and $\hat{\beta}_{-2}$, which fails to reject the null of zero pre-trends at conventional levels. The post-move pooled coefficient

Table F.4. Event-study coefficients, within-physician adaptation around mid-career moves

Event time τ	Estimate	SE	95% CI lo	95% CI hi	p -value
-3	0.063	0.136	-0.20	0.33	0.645
-2	-0.031	0.113	-0.25	0.19	0.785
-1	0				ref.
0	0.443	0.097	0.25	0.63	<0.001
1	0.312	0.114	0.09	0.54	0.006
2	0.184	0.129	-0.07	0.44	0.154
3	0.202	0.128	-0.05	0.45	0.116
4	0.375	0.139	0.10	0.65	0.007
5	0.376	0.163	0.06	0.69	0.021

(constraining $\beta_\tau = \beta$ for $\tau \geq 0$) is 0.350, which we use in the cohort-transition calibration in the main paper.

G Cohort-Transition Simulation

The main paper combines $\hat{\beta}_{\text{train}} = 0.058$ and $\hat{\beta}_{\text{dest}} = 0.350$ into a steady-state multiplier of approximately 1.54. This section reports the derivation of the multiplier and the cohort-transition path implied by gradual cohort replacement.

G.1 Derivation of the Long-Run Multiplier

The multiplier in the main paper rests on three assumptions. First, the destination response operates linearly on the peer mean, so that a one-percentage-point shift in peer-environment cath rate raises own cath rate by β_{dest} percentage points. Second, the system reaches a steady state in which the peer mean equals the average own cath rate across all cardiologists in the HRR. Third, all cohorts contribute equally to the peer mean once the transition is complete. Let $\bar{y}$ denote the steady-state per-cardiologist cath rate and $\Delta\tau$ the uniform training-period cath lab shift. Each cardiologist’s steady-state cath rate equals their direct training imprint plus their adaptation to the peer mean,

$$\bar{y} = \beta_{\text{train}} \cdot \Delta\tau + \beta_{\text{dest}} \cdot \bar{y},$$

which solves to

$$\bar{y} = \frac{\beta_{\text{train}} \cdot \Delta\tau}{1 - \beta_{\text{dest}}}.$$

The multiplier $1/(1 - \beta_{\text{dest}})$ captures the cumulative peer-effect spillover that accumulates as new cohorts displace old ones and incumbents adjust to the evolving peer environment.

G.2 Cohort Transition Path

We trace the per-cardiologist response to a uniform 10-percentage-point increase in training-period cath lab share for all newly entering cohorts, holding the cohort-replacement rate at 5% per year (corresponding to a 20-year career length). Each year, the share of cohorts that received the training shift rises by 5 percentage points until the entire cohort distribution has been replaced at year 20. The cardiologist-level cath rate evolves according to

$$y_t = \beta_{\text{train}} \cdot \Delta\tau \cdot \text{share}_t + \beta_{\text{dest}} \cdot y_{t-1},$$

with the direct training effect compounding through the destination feedback loop. The path approaches the long-run multiplier $1/(1 - \beta_{\text{dest}}) = 1.54$ asymptotically under our within-physician estimate of $\beta_{\text{dest}} = 0.35$. Roughly half of the long-run effect is realized within ten years of the shock, and the system reaches 95% of steady state by year 20.

References

Molitor, David, “The evolution of physician practice styles: evidence from cardiologist migration,” *American Economic Journal: Economic Policy*, 2018, 10 (1), 326–56.